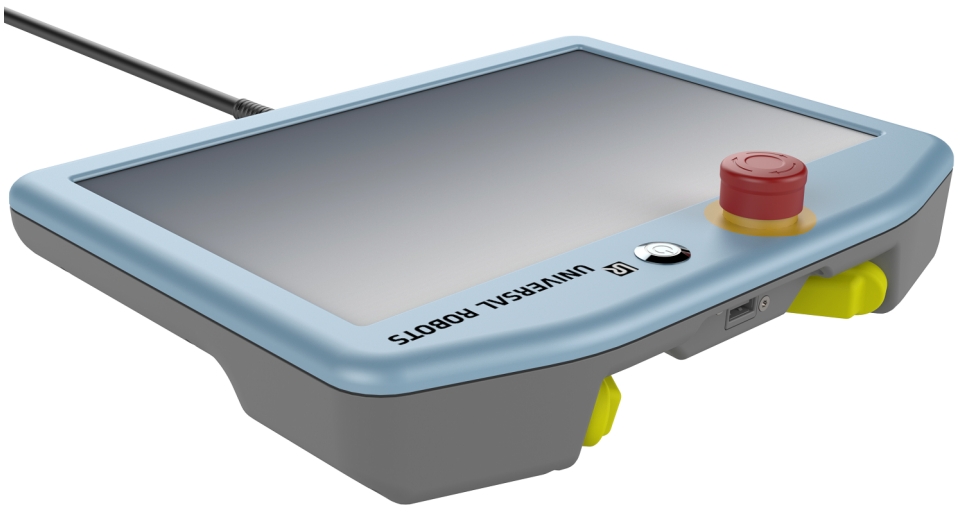
UNIVERSAL ROBOTS



Teach Pendant with 3-Position Enabling Device

Installation Guide

Original instructions (en)

You can download this document in English from the location listed below:

您可以从下面所列位置下载本文档的中文版本:

您可以從下方列出的位置下載本文件的中文版:

Tento dokument si můžete stáhnout v češtině z níže uvedeného umístění:

Du kan downloade dette dokument på dansk fra den placering, der er angivet nedenfor:

Sie können dieses Dokument in deutscher Sprache von dem unten aufgeführten Speicherort herunterladen:

Puede descargar la versión en español de este documento desde la siguiente ubicación:

Vous pouvez télécharger ce document en français depuis l'emplacement ci-dessous :

Ezt a dokumentumot letöltheti magyar nyelven az alább felsorolt helyekről:

È possibile scaricare questo documento in italiano all'indirizzo indicato di seguito:

本文書の日本語版は以下の場所からダウンロードできます。

아래에 나오는 위치에서 이 문서를 한국어로 다운로드할 수 있습니다.

U kunt dit document downloaden in het Nederlands van de hieronder vermelde locatie:

Du kan laste ned dette dokumentet på norsk fra lokasjonen oppført nedenfor:

Wersję tego dokumentu w języku polskim można pobrać z lokalizacji podanej poniżej:

Pode transferir este documento em Português no local listado abaixo:

Puteți descărca acest document în limba română din locația de mai jos:

Загрузить этот документ на русском языке можно по указанному ниже адресу:

Tento dokument v slovenskom jazyku si môžete stiahnuť tu:

Ta dokument lahko v slovenskem jeziku prenesete s spodaj navedenega naslova:

Du kan ladda ned det här dokumentet på svenska från platsen som visas nedanför:

Bu belgeyi aşağıda gösterilen konumdan Türkçe olarak indirebilirsiniz:

https://www.universal-robots.com/download/

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

Contents

1. Introduction **5**

1.1. About this document 5

1.2. What's in the box 5

1.3. Company Details 5

2. Copyright and disclaimers **6**

3. Product description **7**

3.1. Use of Product 8

3.2. Components 8

3.3. Requirements 8

4. Cleaning Your Robot **12**

4.1. Everyday Cleaning 12

4.1.1. Additional Cleaning 12

5. Safety **13**

5.1. Safety Message Types 13

5.2. General safety precautions 14

6. Hardware Installation **15**

7. Software Installation **18**

8. 3PE Teach Pendant button functions **19**

8.1. Using the 3PE buttons 19

8.2. Using Freedrive 20

8.3. Using Move Robot into Position 20

9. Compliance **22**

9.1. Shipping Materials 22

9.2. Certification of Quality 22

9.3. Third Party Certification 22

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

1. Introduction

1.1. About this document

This guide describes how to install a Universal Robots Teach Pendant with 3-Position Enabling (3PE) Device on a Universal Robots e-Series robot and how to configure the software.

The product is also referred to as the 3PE Teach Pendant (3PE TP).

1.2. What's in the box

The box contains the following parts:

- 3PE Teach Pendant
- This guide

1.3. Company Details

Universal Robots A/S

Energivej 51, 5260 Odense S, Denmark

+45 89 93 89 89

https://www.universal-robots.com

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

2. Copyright and disclaimers

Copyright

The information contained herein is the property of Universal Robots A/S and shall not be reproduced in whole or in part without prior written approval of Universal Robots A/S. The information herein is subject to change without notice and should not be construed as a commitment by Universal Robots A/S. This document is periodically reviewed and revised.
Universal Robots A/S assumes no responsibility for any errors or omissions in this document.
Copyright © 2019-2020 by Universal Robots A/S.
The Universal Robots logo is a registered trademark of Universal Robots A/S.

Copyright

NOTICE

Universal Robots continues to improve reliability and performance of its products, and therefore reserves the right to upgrade the product without prior warning. Universal Robots takes care that the content of this document is precise and correct, but takes no responsibility for any errors or missing information.

NOTICE

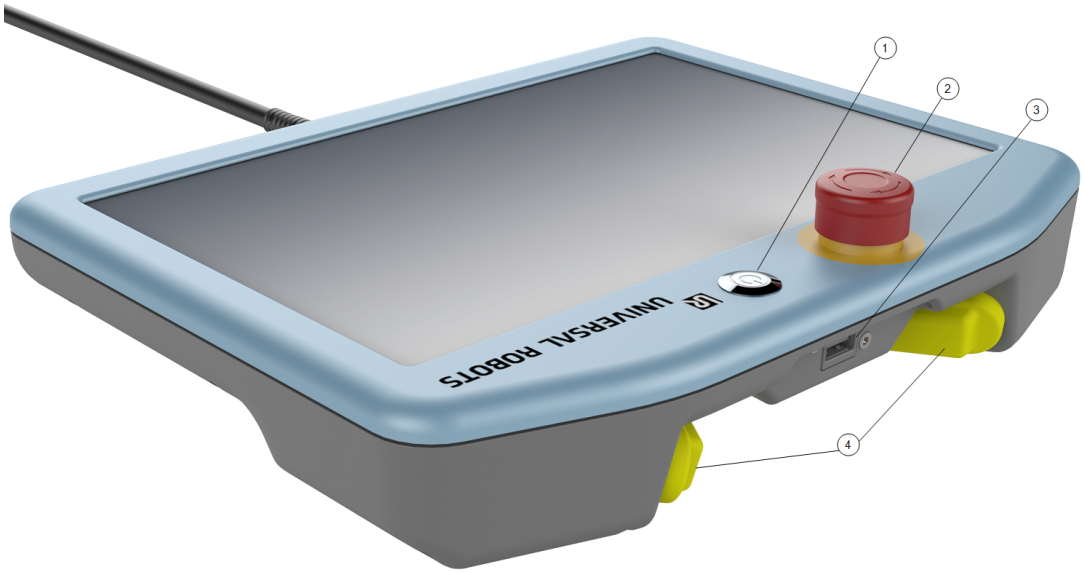
Universal Robots disclaims any liability, even if all guidelines in this document are followed.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

3. Product description

The Teach Pendant with 3-Position Enabling Device is a Universal Robots Teach Pendant with built-in 3-Position Enabling Device functionality.

The enabling buttons are on the underside of the Teach Pendant, as illustrated below. You can use wither button, according to your preference. If the 3PE TP is disconnected, an external 3PE device must be connected and confirmed in the Installation.



1	Power button	2	Emergency Stop button
3	USB port (comes with a dust cover)	4	3PE buttons

A Freedrive robot symbol is located under each 3PE button, as illustrated below.



Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

NOTICE

Failure to install the Teach Pendant with 3-Position Enabling Device properly can result in damage to equipment or property.

- The 3PE Teach Pendant must be installed properly in order to meet regulatory requirements.

3.1. Use of Product

The Teach Pendant with 3-Position Enabling Device is only certified for use with Universal Robots e-Series control boxes.

3.2. Components

The Teach Pendant with 3-Position Enabling Device consists of the following components:

- 3PE Teach Pendant with attached Teach Pendant cable

3.3. Requirements

The robot system software, using the 3PE Teach Pendant must be newer than V5.8.0

NOTICE

Installing an Teach Pendant with 3-Position Enabling Device in a robot system using software version older than V5.8.0 can result in damage to equipment or property.

- Verify your software system number before installing the Teach Pendant with 3-Position Enabling Device.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.



SAFETY NOTICE

UNIVERSAL ROBOTS

To:	**Universal Robots Customers of the UR Teach Pendant with 3PE (3-position enabling device)**
From:	**Universal Robots Global Technical Compliance**
Date:	**20 January 2021**
Subject:	**SAFETY NOTICE Universal Robots e-Series robots with software version 5.9.3 and 5.9.4**

Unexpected motion can occur when a robot application is installed with the following combinations:

- Universal Robots e-Series robot
- UR software versions: 5.9.3 or 5.9.4
- UR Teach Pendant with 3-Position Enabling Device (3PE TP)
- External mode selector switch is connected to the UR Control Box

The risk due to unexpected motion requires all of the following circumstances:

- Automatic mode is selected on the external mode selector switch.
- No protective device is connected to the Safeguard Stop Input of the UR Control Box. (For example: when using a perimeter interlocked guard, light curtain, or safety scanner ect.).
- The operator uses the Teach Pendant to switch to manual mode for teaching or programming.
- The robot is not moving, but it is capable of moving according to its program/programming.
- The operator moves within reach of the robot/end-effector/workpiece, while holding the Teach Pendant.

The probability of unexpected motion happening is extremely low, because all of the above combinations and circumstance are required. As such, note the following:

- All safety functions are active and functioning properly. This includes the 3PE which is not active when the robot control is in automatic mode.
- The operator is typically more attentive to robot activity due to planned teaching or programming tasks.
- Operator attention is more likely to be directed to the robot application; it is unlikely that the incorrect Teach Pendant mode indication will be noticed.

To eliminate the risks associated with this issue, update to software version 5.10 or higher. The robot application can remain operational before updating the software ONLY if the external mode selector switch is disconnected from the UR Control Box. With 5.9.3 and 5.9.4 after removing the external mode selector, train personnel to select the mode using the UR Teach Pendant with 3PE.
Software version 5.10 or higher eliminates this issue and Teach Pendant with 3PE owners are urged to update.
As always, proper inspection and maintenance is required as stated in the UR Service Manual.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

How to recognize the Teach Pendant with 3-Position Enabling Device (3PE TP)
Only the UR e-Series robots with software 5.9.3 or 5.9.4 and a 3PE TP are affected if an external mode selector is used.
The 3PE TP only functions with UR e-Series robots.

Helpful links:

- 3PE TP support article
- SW 5.10 release notes
- How to update software: Service Manual, Chapter 5
- 3PE TP Installation Guide



If you have any questions about this Safety Notice, you can email questions to: **SafetyNotice@Universal-Robots.com** It will be directed to the appropriate department.
You can directly contact your regional Universal Robots representative at: https://www.universal-robots.com/about-universal-robots/contact-info.

EN Original

zh-CN

zh-TW

cs

da

de

es

fr

no

it

ja

sv

ko

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

nl

hu

pl

pt

ro

ru

sk

sl

tr

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

4. Cleaning Your Robot

4.1. Everyday Cleaning

You can wipe away any dust/dirt/oil observed on the robot arm and/or Teach Pendant using a cloth and one of the following cleaning agents: Water, 70% Isopropyl alcohol, 10% Ethanol alcohol or 10% Naphtha. In rare cases, very small amounts of grease can be visible from the joint. This does not affect the function, use, or lifetime of the joint.

4.1.1. Additional Cleaning

Due to the added focus on cleaning your robot, Universal Robots recommends cleaning with 70% isopropyl alcohol (rubbing alcohol).

1. Wipe the robot with a hard twisted micro fiber cloth and 70% isopropyl alcohol (rubbing alcohol).
2. Let the 70% isopropyl alcohol dwell on the robot for 5 minutes, and then clean the robot using standard cleaning procedure.

DO NOT USE BLEACH. Do not use bleach in any diluted cleaning solution.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

5. Safety

5.1. Safety Message Types

Description

Safety messages are used to emphasize important information. Read all the messages to help ensure safety and to prevent injury to personnel and product damage.



WARNING

Indicates a hazardous situation that, if not avoided, can result in death or serious injury.



WARNING: ELECTRICITY

Indicates a hazardous electrical situation that, if not avoided, can result in death or serious injury.



WARNING: HOT SURFACE

Indicates a hazardous hot surface where injury can result from contact and non-contact proximity.



CAUTION

Indicates a hazardous situation that, if not avoided, can result in injury.



GROUND

Indicates grounding.



PROTECTIVE GROUND

Indicates protective grounding.



NOTICE

Indicates the risk of damage to equipment and/or information to be noted.



READ MANUAL

Indicates more detailed information that should be consulted in the manual.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

5.2. General safety precautions

Read the general safety precautions before installing the 3PE TP.

For more information on safety, refer to the Safety chapter in the robot User Manual.



WARNING: ELECTRICITY

Installing or maintaining equipment connected to a power source can lead to electric shock.

- Disconnect the equipment from the power source before installation or maintenance.



WARNING

Incorrect connection of the power source or ground wires can result in equipment damage or personnel injury.
Damage caused by invalid power source connection is not covered by warranty. Before starting the operation:

- Ensure that the power source wiring is correct.
- Ensure that the grounding is correct.



WARNING

Failure to verify and validate safeguarding and functionality can result in death or serious injury.

- Ensure all risk reduction works as intended and achieves the needed risk reduction.



WARNING

Failure to perform a risk assessment before installation and operation can result in personnel injury or equipment damage.

- Perform a risk assessment before installation and operation.
- Read the UR User Manual and UR Service Manual.



NOTICE

This product is designed for use as part of the Universal Robots e-Series robot. General safety considerations that are valid for the Universal Robots e-Series robot are also valid for this product.

- For more information on safety, refer to the Safety section in the Universal Robots e-Series robot User Manual.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

6. Hardware Installation

Description

UR20 and UR30 robots have a built-it 3 Position Enabling functionality.
e-Series robots do not have built-in 3 Position Enabling functionality. You can use the following instructions to replace a Teach Pendant and install a 3 Position Enabling Teach Pendant (3PE TP).

Required items

You need the following items to install the 3PE Teach Pendant:

- 3PE Teach Pendant

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

To remove a Teach Pendant

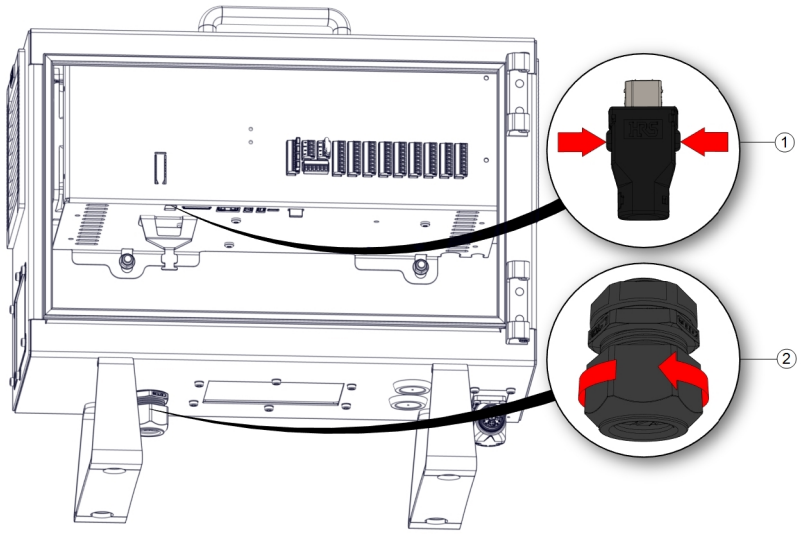
NOTICE

Replacing the Teach Pendant can result in the system reporting a fault on start-up.

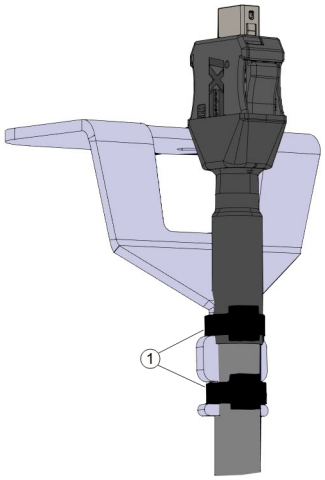
- Always select the correct configuration for the type of Teach Pendant.

To remove the standard Teach Pendant:

1. Power down the control box and disconnect the main power cable from the power source.
2. Remove and discard the two cable ties used for mounting the Teach Pendant cables.
3. Press in the clips on both sides of the Teach Pendant plug as illustrated, and pull down to disconnect from the Teach Pendant port.
4. Fully open/loosen the plastic grommet at the bottom of the control box and remove the Teach Pendant plug and cable.
5. Gently remove the Teach Pendant cable and Teach Pendant.



1	Clips	2	Plastic grommet



1	Cable ties

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

To install a 3PE Teach Pendant

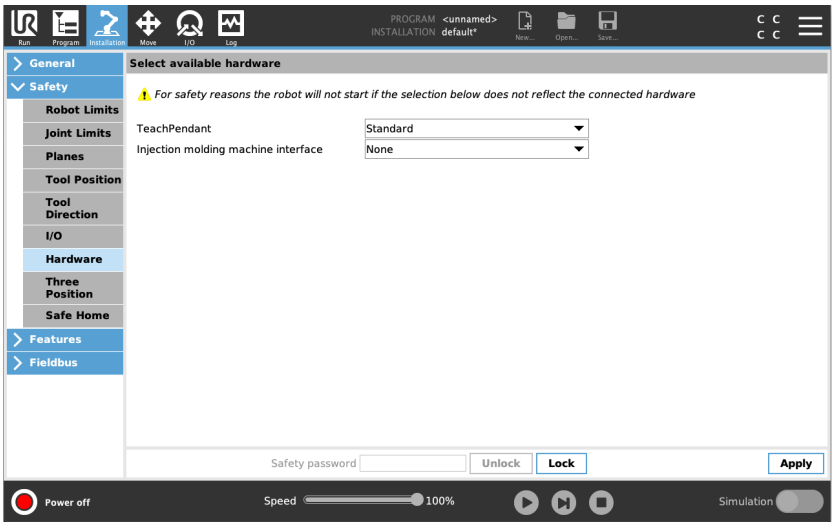
1. Place the Teach Pendant plug and cable in through the bottom of the control box and fully close/tighten the plastic grommet.
2. Push the Teach Pendant plug into the Teach Pendant port to connect.
3. Use two new cable ties to mount the Teach Pendant cables.
4. Connect the main power cable to the power source and power on the control box.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

7. Software Installation

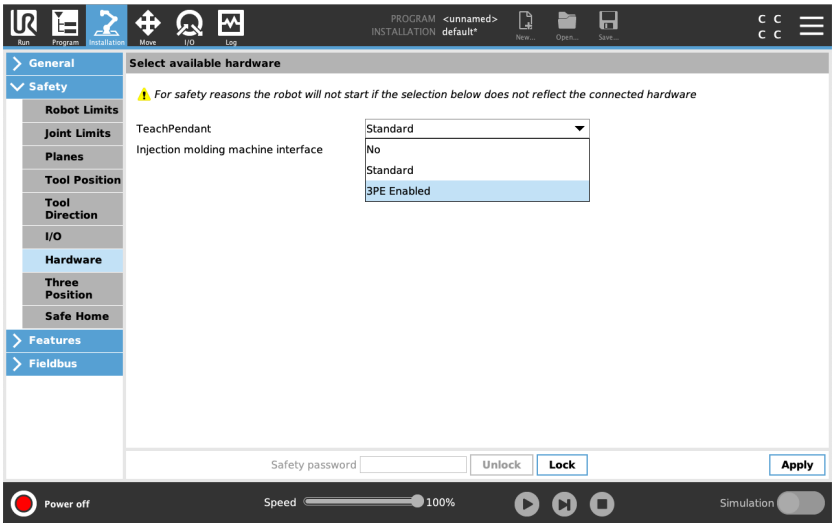
To configure the 3PE TP software

1. On PolyScope, in the Header, tap Installation and select **Safety**.



2. Tap Hardware and unlock the options on the **Select available hardware** screen.

   A password is required to unlock this screen.



3. In the **Teach Pendant** drop-down list, select **3PE Enabled**.
4. Tap **Apply** to restart the system. PolyScope continues to run.
5. Tap **Confirm Safety Configuration** to complete the 3PE Teach Pendant software installation.
6. As the robot restarts and initializes, light-press the 3PE button and tap **Start** on PolyScope.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

8. 3PE Teach Pendant button functions

This section describes the functions of the 3-Position Enabling Device buttons on the 3PE Teach Pendant.

8.1. Using the 3PE buttons

NOTICE

The 3PE buttons are only active in Manual mode. In Automatic mode, robot movement does not require 3PE button action.

The following table contains the available functions of the 3PE buttons.

Position		Description	Action
1	Release	There is no pressure on the 3PE button. It is not pressed.	Robot movement is stopped in Manual mode. Power is not removed from the robot arm and the brakes remain released.
2	Light-press (Grip lightly)	There is some pressure on the 3PE button. It is pressed to a middle point.	Allows your program to play when the robot is in Manual mode.
3	Tight-press (Grip tightly)	There is full pressure on the 3PE button. It is pressed all the way down.	Robot movement is stopped in Manual mode. Robot is in 3PE Stop.

To play a program:

1. On PolyScope, ensure the robot is set to **Manual mode**, or switch to **Manual mode**.
2. Maintain a light-press on the 3PE button.
3. On PolyScope, tap **Play** to run the program.

   The program runs if the robot arm is in the first position of the program.

   If the robot is not in the first position of the program, the **Move Robot into Position screen** appears.

To stop a program:

1. Release the 3PE button or, on PolyScope, tap **Stop**.

To pause a program:

1. Release the 3PE button, or, on PolyScope, tap **Pause**.

   To continue the program execution, keep the 3PE button light pressed and tap **Resume** on PolyScope.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

8.2. Using Freedrive

Freedrive allows the robot arm to be manually pulled into desired positions and/or poses. For more information, refer to the Freedrive section in the robot User Manual.

Description	Freedrive allows the robot arm to be manually pulled into desired positions and/or poses.
Use the 3PE button to freedrive the robot arm	Rapidly light-press, release, light-press again and keep holding the 3PE button in this position. Now you can pull the robot arm into a desired position, while the light-press is maintained.

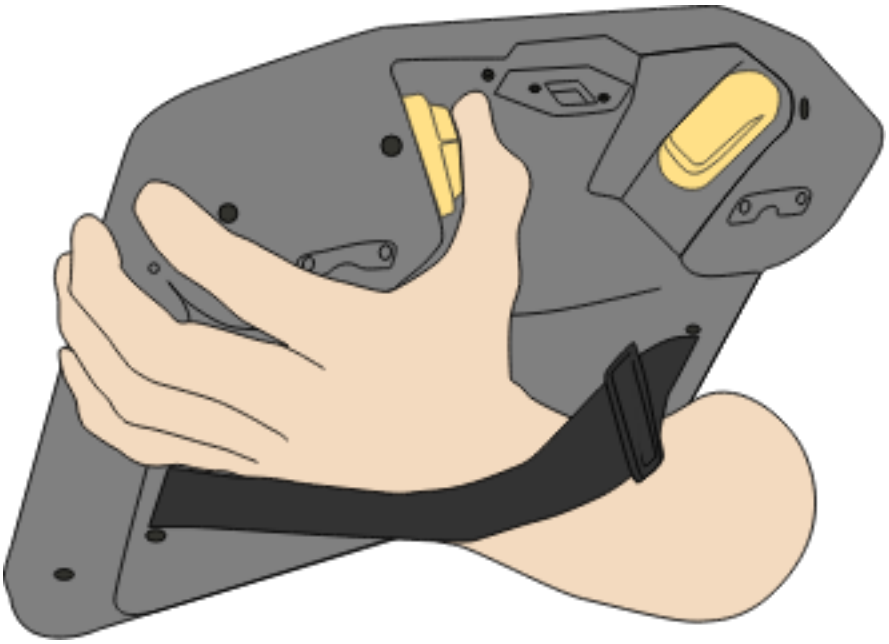
8.3. Using Move Robot into Position

Move Robot into Position allows the robot arm to move to that start position, after you complete a program. The robot arm must be in the start position before you can run the program.

For more information, refer to the Move Robot into Position section in the robot User Manual.

To use the 3PE button to move the robot arm into position:

1. When your program is complete, press **Play**
2. Select **Play from beginning**.
   On PolyScope, the **Move Robot into Position** screen appears displaying robot arm movement.
3. Light-press and hold the 3PE button.

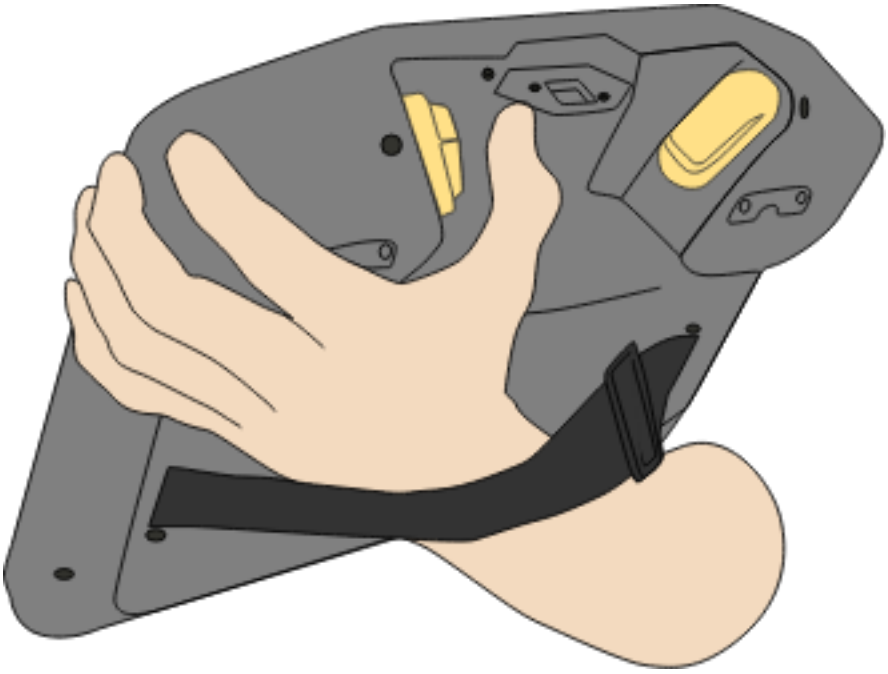


4. Now, on PolyScope, press and hold Automove, for the robot arm to move to the start position.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

The Play Program screen appears.

5. Maintain a light-press on the 3PE button to run your program.
6. Release the 3PE button to stop your program.



Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

9. Compliance

The recommendations, certificates and declarations, below, are valid for Universal Robots Teach Pendant with integrated 3-position enabling (3PE) device.

9.1. Shipping Materials

As stated by our suppliers, Universal Robots e-Series robots shipping materials comply with the ISPM-15 requirements for producing wood packaging material and are marked accordingly.

9.2. Certification of Quality

Universal Robots is certified compliant with the standard ISO 9001:2015 by Bureau Veritas.

Universal Robots e-Series robots undergo continuous internal testing and end-of-line test procedures. UR testing processes undergo continuous review and improvement.

9.3. Third Party Certification

Universal Robots Teach Pendant with 3-Position Enabling (3PE) Device is certified compliant with the standard EN ISO 10218-1.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

Copyright © 2019-2020 by Universal Robots A/S. All rights reserved.

Software Name: PolyScope 5
Software Version: 5.25
Document Version: 10.16.265